

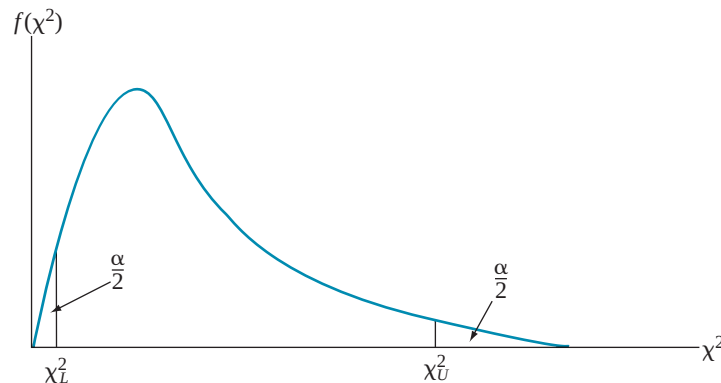
under the curve, we compute $a = 1 - .025$ and obtain 5.629 from Table 7 in the Appendix. Combining these two values, we have that the area under the curve between 5.629 and 26.12 is $1 - .025 - .025 = .95$. (See Figure 7.4.) We can use this information to form a confidence interval for σ^2 . Because the chi-square distribution is not symmetrical, the confidence intervals based on this distribution do not have the usual form, estimate $\pm$ error, as we saw for μ and $\mu_1 - \mu_2$. The $100(1 - \alpha)\%$ confidence interval for σ^2 is obtained by dividing the estimator of σ^2 , s^2 , by the lower and upper $\alpha/2$ percentiles, χ_L^2 and χ_U^2 , as described here.

General Confidence Interval for σ^2 (or σ) with Confidence Coefficient $(1 - \alpha)$

$$\frac{(n-1)s^2}{\chi_U^2} < \sigma^2 < \frac{(n-1)s^2}{\chi_L^2}$$

where χ_U^2 is the upper-tail value of chi-square for $df = n - 1$ with area $\alpha/2$ to its right, and χ_L^2 is the lower-tail value with area $\alpha/2$ to its left (see Figure 7.5). We can determine χ_U^2 and χ_L^2 for a specific value of df by obtaining the critical value in Table 7 of the Appendix corresponding to $a = \alpha/2$ and $a = 1 - \alpha/2$, respectively. (Note: The confidence interval for σ is found by taking square roots throughout.)

FIGURE 7.5
Upper-tail and lower-tail values of chi-square



EXAMPLE 7.1

The machine that fills 500-gram coffee containers for a large food processor is monitored by the quality control department. Ideally, the amount of coffee in a container should vary only slightly about the nominal 500-gram value. If the variation was large, then a large proportion of the containers would be either underfilled, thus cheating the customer, or overfilled, thus resulting in economic loss to the company. The machine was designed so that the weights of the 500-gram containers would have a normal distribution with mean value of 506.6 grams and a standard deviation of 4 grams. This would produce a population of containers in which at most 5% of the containers weighed less than 500 grams. To maintain a population in which at most 5% of the containers are underweight, a random sample of 30 containers is selected every hour. These data are then used to determine whether the mean and standard deviation are maintained at their nominal values. The weights from one of the hourly samples are given here.

501.4	498.0	498.6	499.2	495.2	501.4	509.5	494.9	498.6	497.6
505.5	505.1	499.8	502.4	497.0	504.3	499.7	497.9	496.5	498.9
504.9	503.2	503.0	502.6	496.8	498.2	500.1	497.9	502.2	503.2

Estimate the mean and standard deviation in the weights of coffee containers filled during the hour, in which the random sample of 30 containers was selected using a 99% confidence interval.

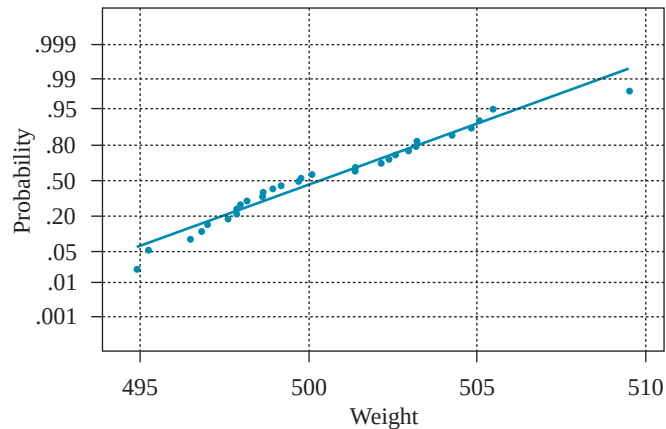
Solution For these data, we find

$$\bar{y} = 500.453 \quad \text{and} \quad s = 3.433$$

To use our method for constructing a confidence interval for μ and σ , we must first check whether the weights are a random sample from a normal population. Figure 7.6 is a normal probability plot of the 30 weights. The 30 values fall near the straight line. Thus, the normality condition appears to be satisfied. The confidence coefficient for this example is $1 - \alpha = .99$. The upper-tail chi-square value can be obtained from Table 7 in the Appendix, for $df = n - 1 = 29$ and $\alpha = \alpha/2 = .005$. Similarly, the lower-tail chi-square value is obtained from Table 7, with $\alpha = 1 - \alpha/2 = .995$. Thus,

$$\chi_L^2 = 13.12 \quad \text{and} \quad \chi_U^2 = 52.34$$

FIGURE 7.6
Normal probability plot of
container weights



The 99% confidence interval for σ is then

$$\sqrt{\frac{29(3.433)^2}{52.34}} < \sigma < \sqrt{\frac{29(3.433)^2}{13.12}}$$

or

$$2.56 < \sigma < 5.10$$

Thus, we are 99% confident that the standard deviation in the weights of coffee cans lies between 2.56 and 5.10 grams. The designed value for σ , 4 grams, falls within our confidence interval. Using our results from Chapter 5, a 99% confidence interval for μ is

$$500.453 \pm 2.756 \frac{3.433}{\sqrt{30}} \quad 500.453 \pm 1.73$$

or

$$498.7 < \mu < 502.2$$

Thus, it appears the machine is underfilling the containers, because 506.6 grams does not fall within the confidence limits.

In addition to estimating a population variance, we can construct a statistical test of the null hypothesis that σ^2 equals a specified value, σ_0^2 . This test procedure is summarized here.

**Statistical Test for σ^2
(or σ)**

$$H_0: \begin{array}{l} 1. \sigma^2 \leq \sigma_0^2 \\ 2. \sigma^2 \geq \sigma_0^2 \\ 3. \sigma^2 = \sigma_0^2 \end{array} \quad H_a: \begin{array}{l} 1. \sigma^2 > \sigma_0^2 \\ 2. \sigma^2 < \sigma_0^2 \\ 3. \sigma^2 \neq \sigma_0^2 \end{array}$$

$$\text{T.S.: } \chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$$

R.R: For a specified value of α ,

1. Reject H_0 if χ^2 is greater than χ_{α}^2 , the upper-tail value for $a = \alpha$ and $\text{df} = n - 1$.
2. Reject H_0 if χ^2 is less than $\chi_{1-\alpha}^2$, the lower-tail value for $a = 1 - \alpha$ and $\text{df} = n - 1$.
3. Reject H_0 if χ^2 is greater than $\chi_{\alpha/2}^2$, based on $a = \alpha/2$ and $\text{df} = n - 1$, or less than $\chi_{1-\alpha/2}^2$, based on $a = 1 - \alpha/2$ and $\text{df} = n - 1$.

Check assumptions and draw conclusions.

EXAMPLE 7.2

A manufacturer of a specific pesticide useful in the control of household bugs claims that its product retains most of its potency for a period of at least 6 months. More specifically, it claims that the drop in potency from 0 to 6 months will vary in the interval from 0% to 8%. To test the manufacturer's claim, a consumer group obtained a random sample of 20 containers of pesticide from the manufacturer. Each can was tested for potency and then stored for a period of 6 months at room temperature. After the storage period, each can was again tested for potency. The percentage drop in potency was recorded for each can and is given here.

0.5 3.5 4.4 6.0 6.6 5.4 7.9 4.6 5.4 5.7
2.5 1.1 5.9 2.7 2.3 1.4 1.8 5.8 0.2 7.1

Use these data to determine whether there is sufficient evidence to indicate that the population of potency drops has more variability than claimed by the manufacturer. Use $\alpha = .05$.

Solution The manufacturer claimed that the population of potency reductions has a range of 8%. Dividing the range by 4, we obtain an approximate population standard deviation of $\sigma = 2\%$ (or $\sigma^2 = 4$).

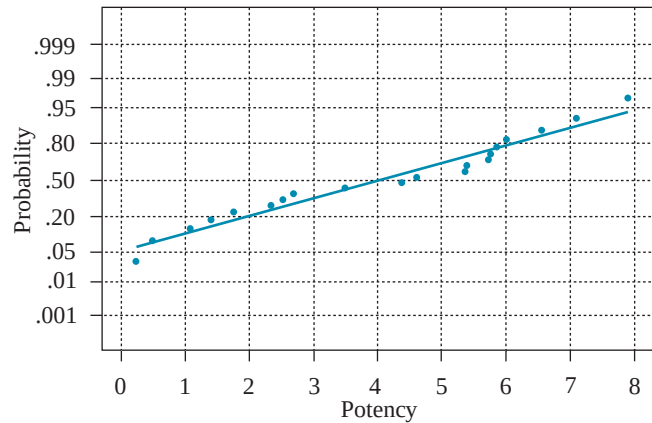
The appropriate null and alternative hypotheses are

$$H_0: \sigma^2 \leq 4 \text{ (i.e., we assume the manufacturer's claim is correct)}$$

$$H_a: \sigma^2 > 4 \text{ (i.e., there is more variability than claimed by the manufacturer)}$$

To use our inference techniques for variances, we must first check the normality of the data. From Figure 7.7 we observe that the plotted points fall nearly on the

FIGURE 7.7
Normal probability plot for
potency data



straight line. Thus, the normality condition appears to be satisfied. From the 20 data values, we compute $s^2 = 5.45$. The test statistic and rejection region are as follows:

$$\text{T.S.: } \chi^2 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{19(5.45)}{4} = 25.88$$

R.R.: For $\alpha = .05$, we will reject H_0 if the computed value of chi-square is greater than 30.14, obtained from Table 7 in the Appendix for $\alpha = .05$ and $df = 19$.

Conclusion: Since the computed value of chi-square, 25.88, is less than the critical value, 30.14, there is insufficient evidence to reject the manufacturer's claim, based on $\alpha = .05$. The p -value $= P(\chi_{19}^2 > 25.88) = .14$ can be found using a computer program. Using Table 7 in the Appendix, we can only conclude that p -value $> .10$, because the p -value $= P(\chi_{19}^2 > 25.88) > P(\chi_{19}^2 > 27.20) = .10$. The sample size is relatively small and the p -value is only moderately large, so the consumer group is not prepared to accept $H_0: \sigma^2 \leq 4$. Rather, it would be wise to do additional testing with a larger sample size before reaching a definite conclusion.

The inference methods about σ are based on the condition that the random sample is selected from a population having a normal distribution similar to the requirements for using t distribution-based inference procedures. However, when sample sizes are moderate to large ($n \geq 30$), the t distribution-based procedures can be used to make inferences about μ even when the normality condition does not hold, because for moderate to large sample sizes the Central Limit Theorem provides that the sampling distribution of the sample mean is approximately normal. Unfortunately, the same type of result does not hold for the chi-square-based procedures for making inferences about σ ; that is, if the population distribution is distinctly nonnormal, then these procedures for σ are not appropriate even if the sample size is large. Population nonnormality, in the form of skewness or heavy tails, can have serious effects on the nominal significance and confidence probabilities for σ . If a boxplot or normal probability plot of the sample data shows substantial skewness or a substantial number of outliers, the chi-square-based inference procedures should not be applied. There are some alternative approaches that involve computationally elaborate inference procedures. One such procedure is the bootstrap. Bootstrapping is a technique that provides a simple and practical way to estimate the uncertainty in sample statistics like